Chapter 2: Configuring a Network Operating System



Introduction to Networks

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Chapter 2

2.0 Introduction

2.1 IOS Bootcamp

2.2 Getting Basic

2.3 Addressing Schemes

2.4 Summary

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2.1 IOS Bootcamp

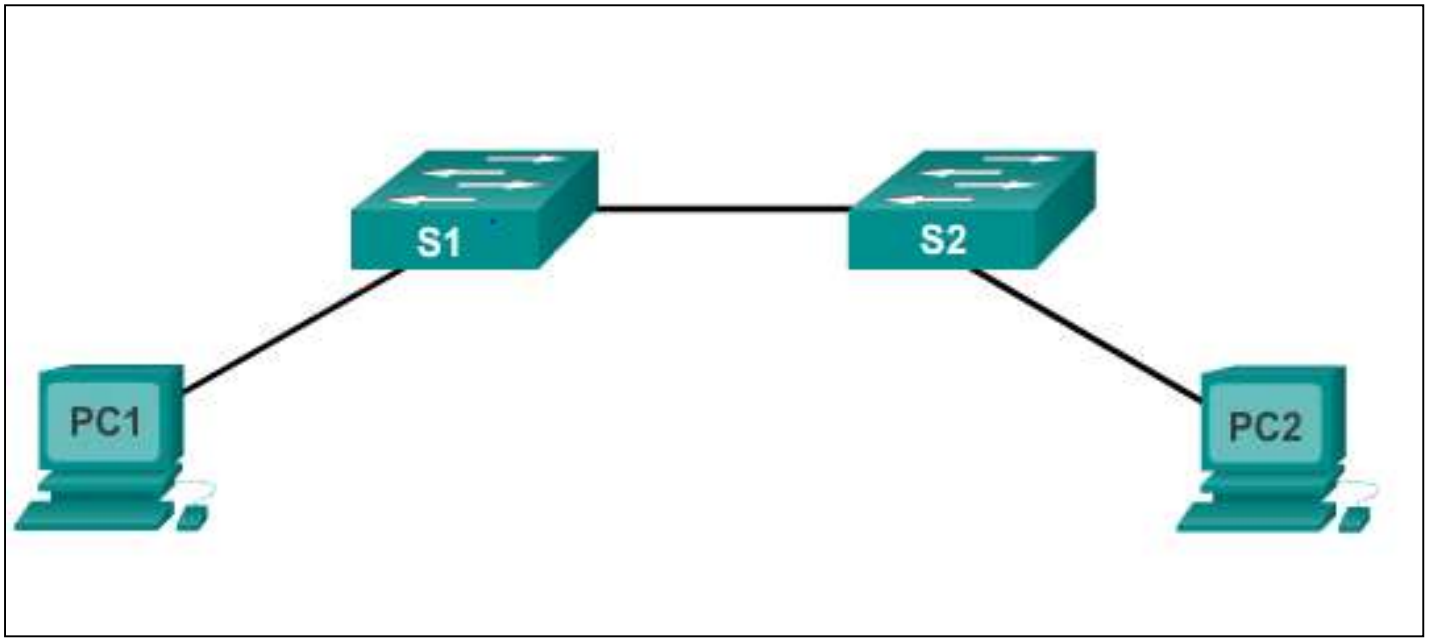


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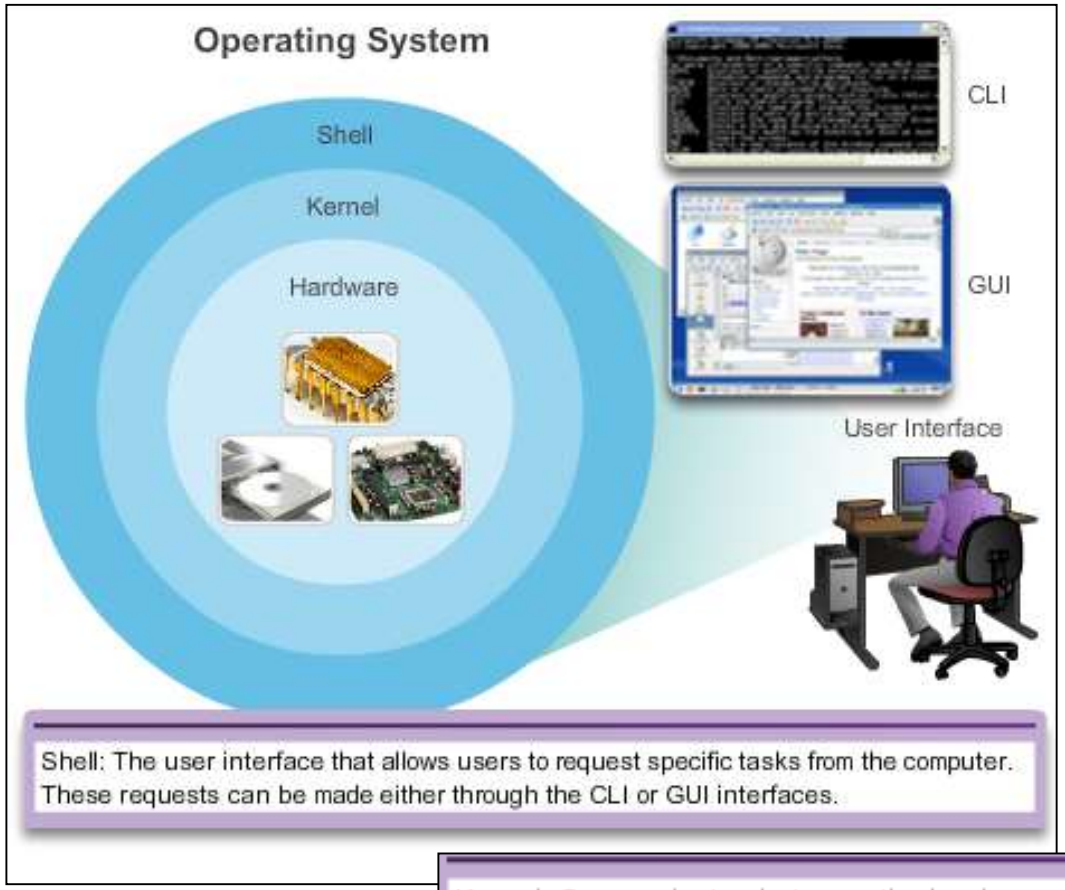
Operating Systems

- All networking equipment dependent on operating systems
- The operating system on home routers is usually called **firmware**
- Cisco IOS – Collection of network operating systems used on Cisco devices



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Operating Systems (cont.)



Shell: The user interface that allows users to request specific tasks from the computer. These requests can be made either through the CLI or GUI interfaces.

Kernel: Communicates between the hardware and software of a computer and manages how hardware resources are used to meet software requirements.

Hardware: The physical part of a computer including underlying electronics.

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Purpose of OS

- PC operating systems (Windows 8 and OS X) perform technical functions that enable:
  - Use of a mouse
  - View output
  - Enter text
- Switch or router IOS provides options to:
  - Configure interfaces
  - Enable routing and switching functions
- All networking devices come with a default IOS
- Possible to upgrade the IOS version or feature set
- In this course, primary focus is Cisco IOS Release 15.x

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Location of the Cisco IOS

Cisco IOS stored in **Flash**

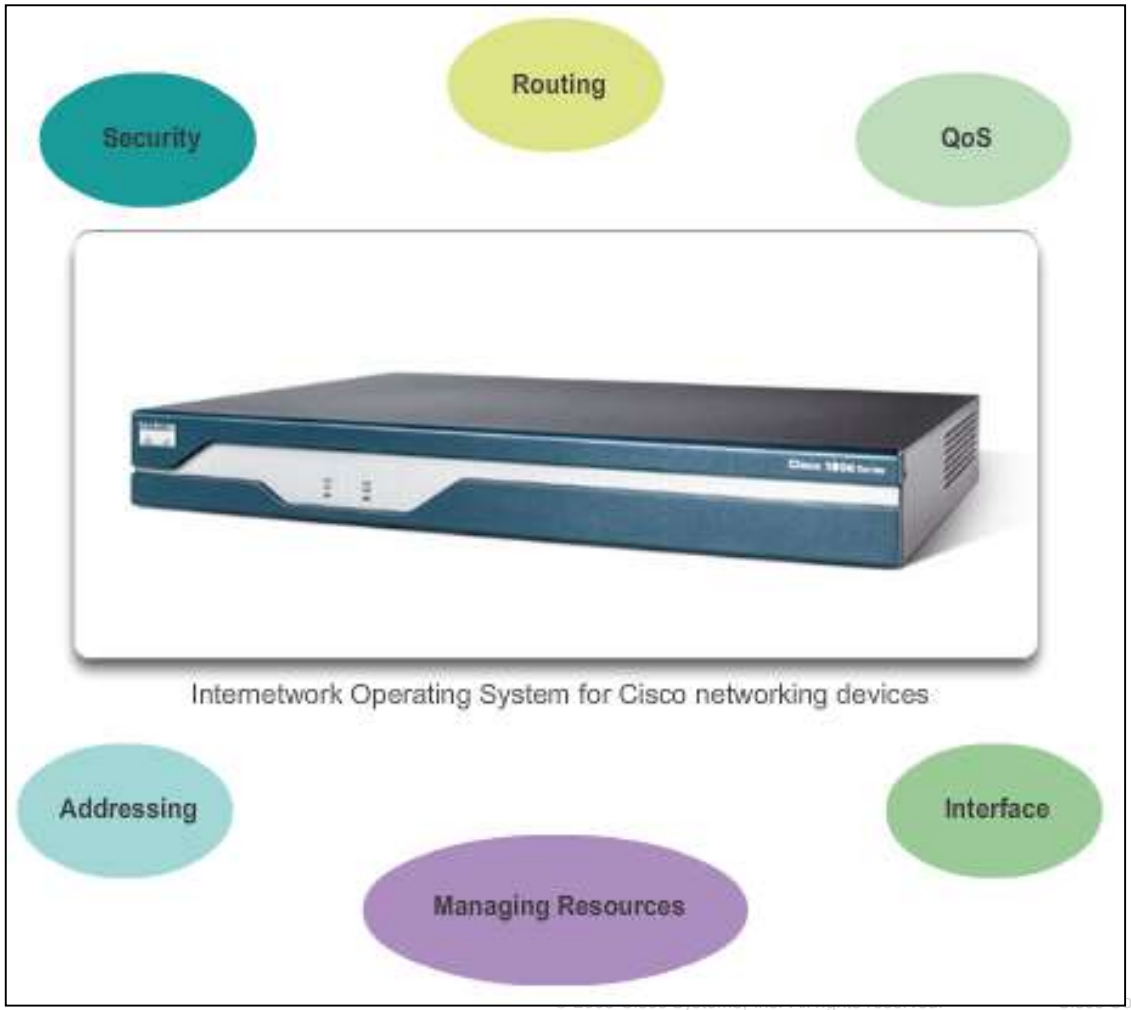
- Non-volatile storage, not lost when power is lost
- Can be changed or overwritten as needed
- Can be used to store multiple versions of IOS
- IOS copied from flash to volatile RAM
- Quantity of flash and RAM memory determines IOS that can be used



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IOS Functions

These are the major functions performed or enabled by Cisco routers and switches.



Console Access Method

Most common methods to access the CLI:

- Console
- Telnet or SSH
- AUX port



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Console Access Method

Console Port

- Device is accessible even if no networking services have been configured (out-of-band)
- Need a special console cable
- Allows configuration commands to be entered
- Should be configured with passwords to prevent unauthorized access
- Device should be located in a secure room so console port cannot be easily accessed



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Telnet, SSH, and AUX Access Methods

Telnet

- Method for remotely accessing the CLI over a network
- Require active networking services and one active interface that is configured

Secure Shell (SSH)

- Remote login similar to Telnet, but utilizes more security
- Stronger password authentication
- Uses encryption when transporting data

Aux Port

- Out-of-band connection
- Uses telephone line
- Can be used like console port

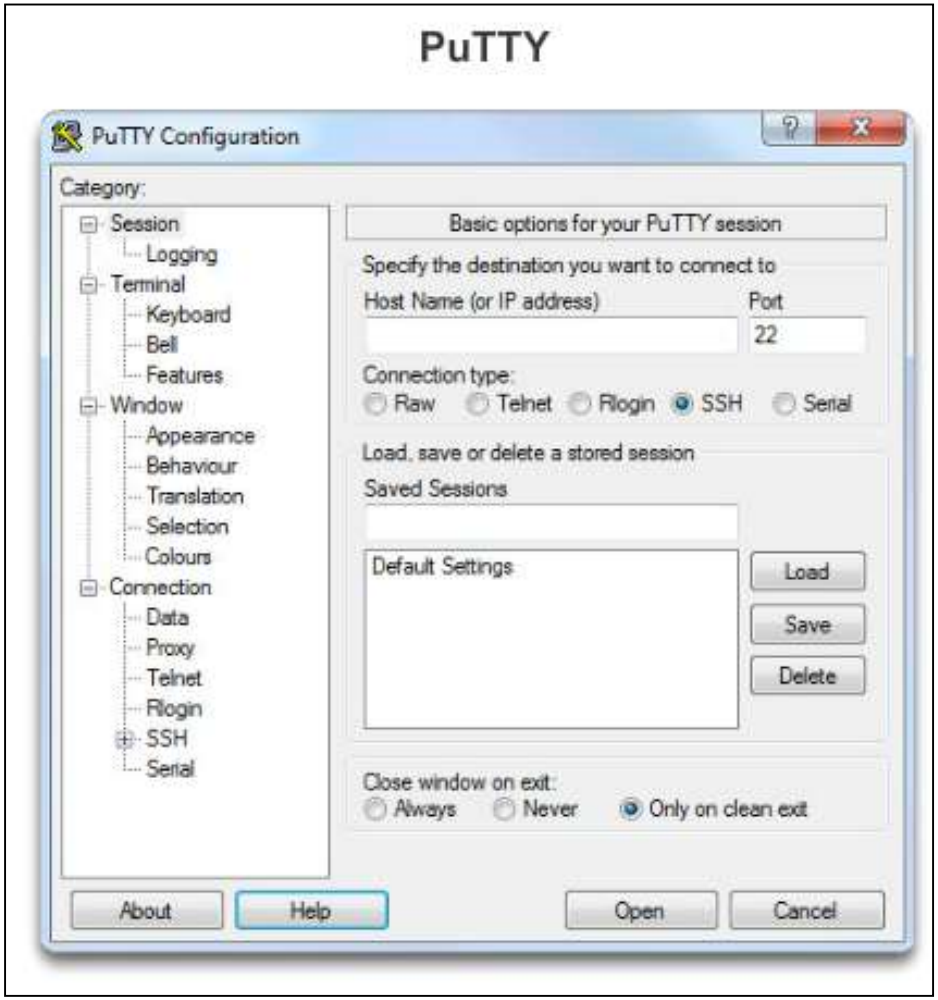


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Terminal Emulation Programs

Software available for connecting to a networking device:

- PuTTY
- Tera Term
- SecureCRT
- HyperTerminal
- OS X Terminal

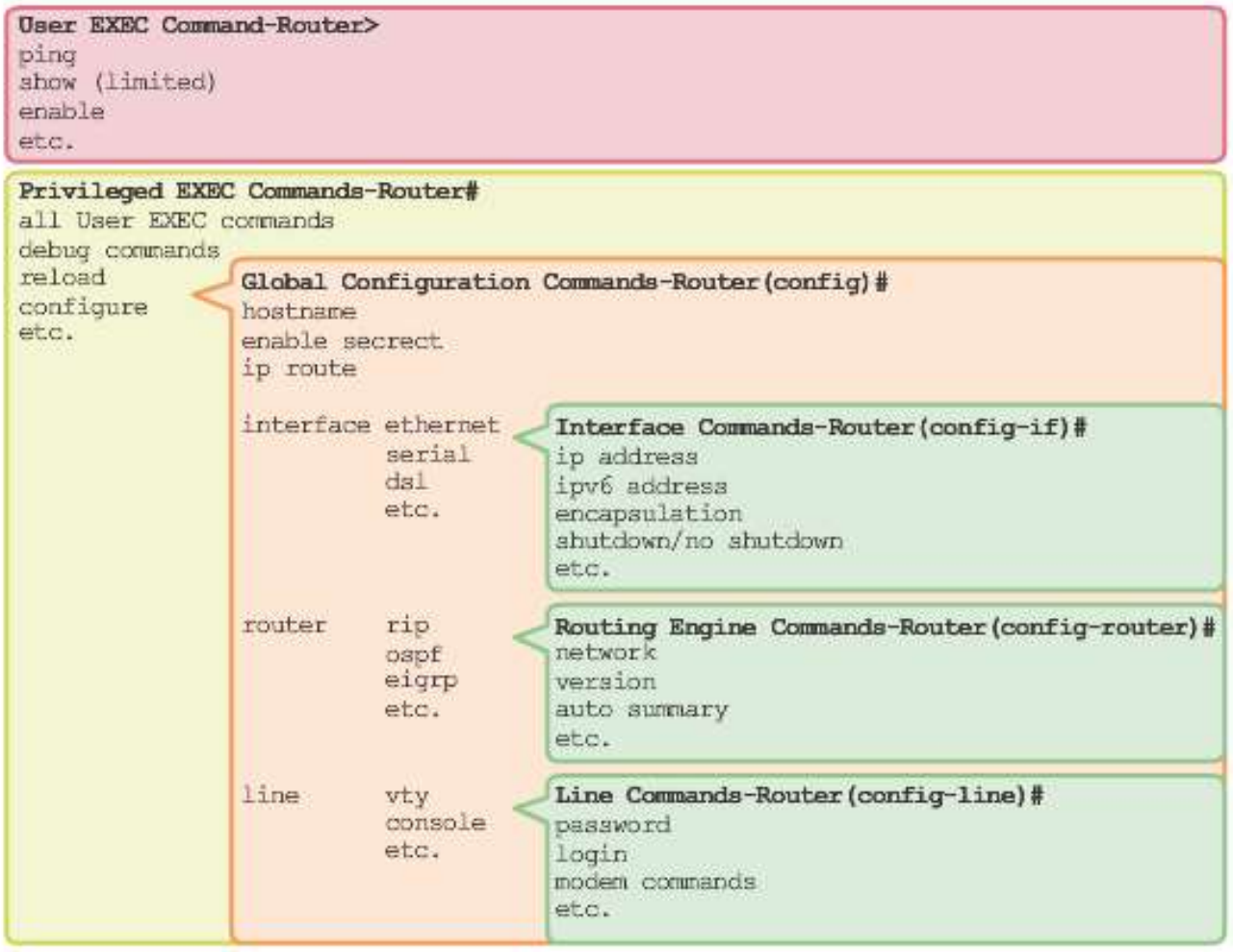


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Navigating the IOS

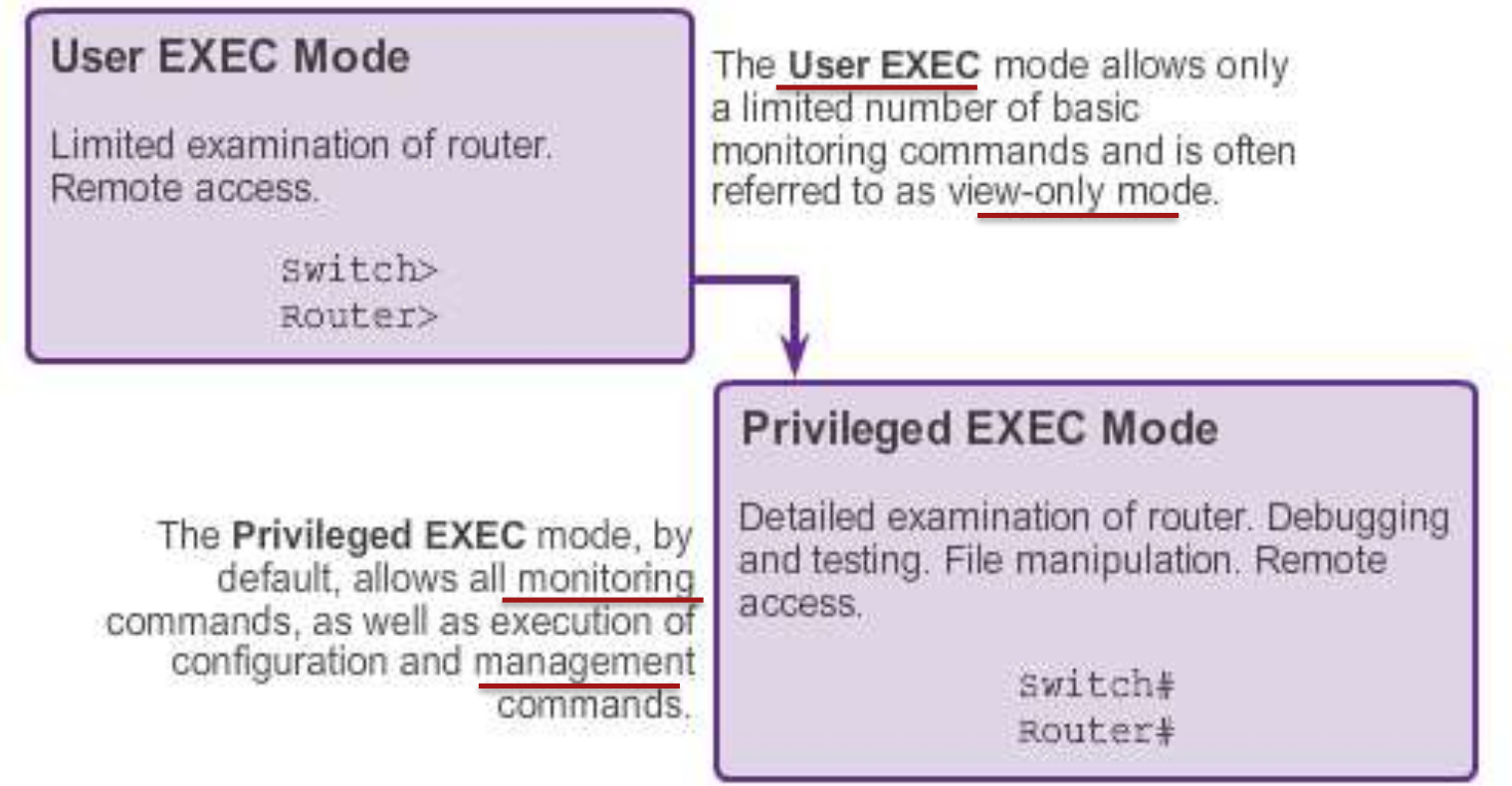
Cisco IOS Modes of Operation

IOS Mode Hierarchical Structure



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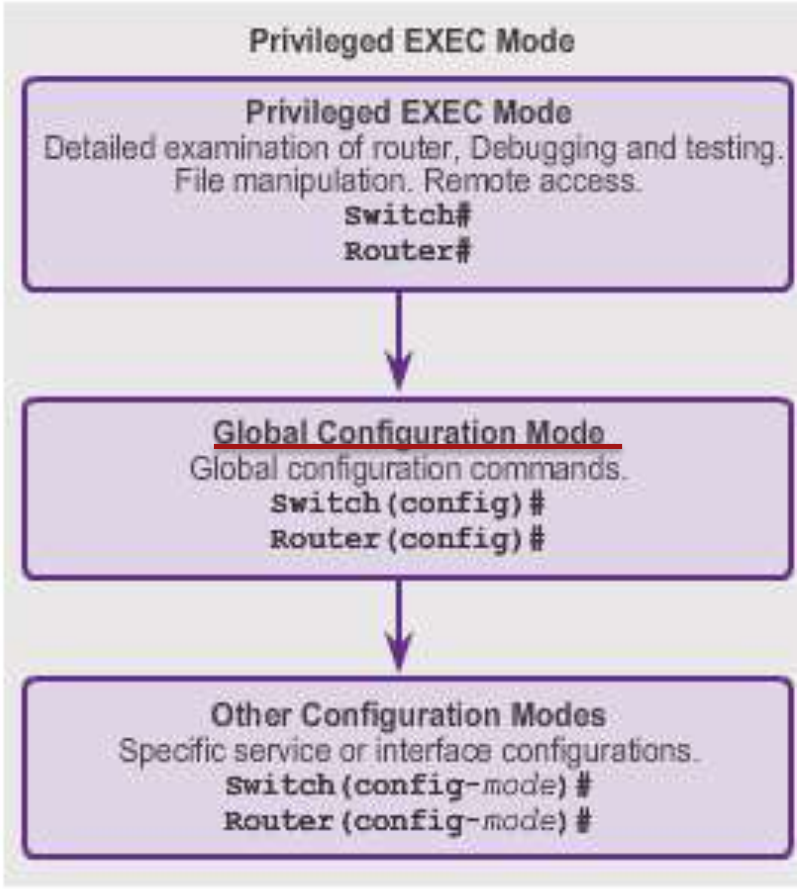
Primary Modes



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Navigating the IOS

Global Configuration Mode and Submodes



IOS Prompt Structure

```
Router>ping 192.168.10.5

Router#show running-config

Router(config)#Interface FastEthernet 0/0

Router(config-if)#ip address 192.168.10.1 255.255.255.0
```

The prompt changes to denote the current CLI mode.

```
Switch>ping 192.168.10.9

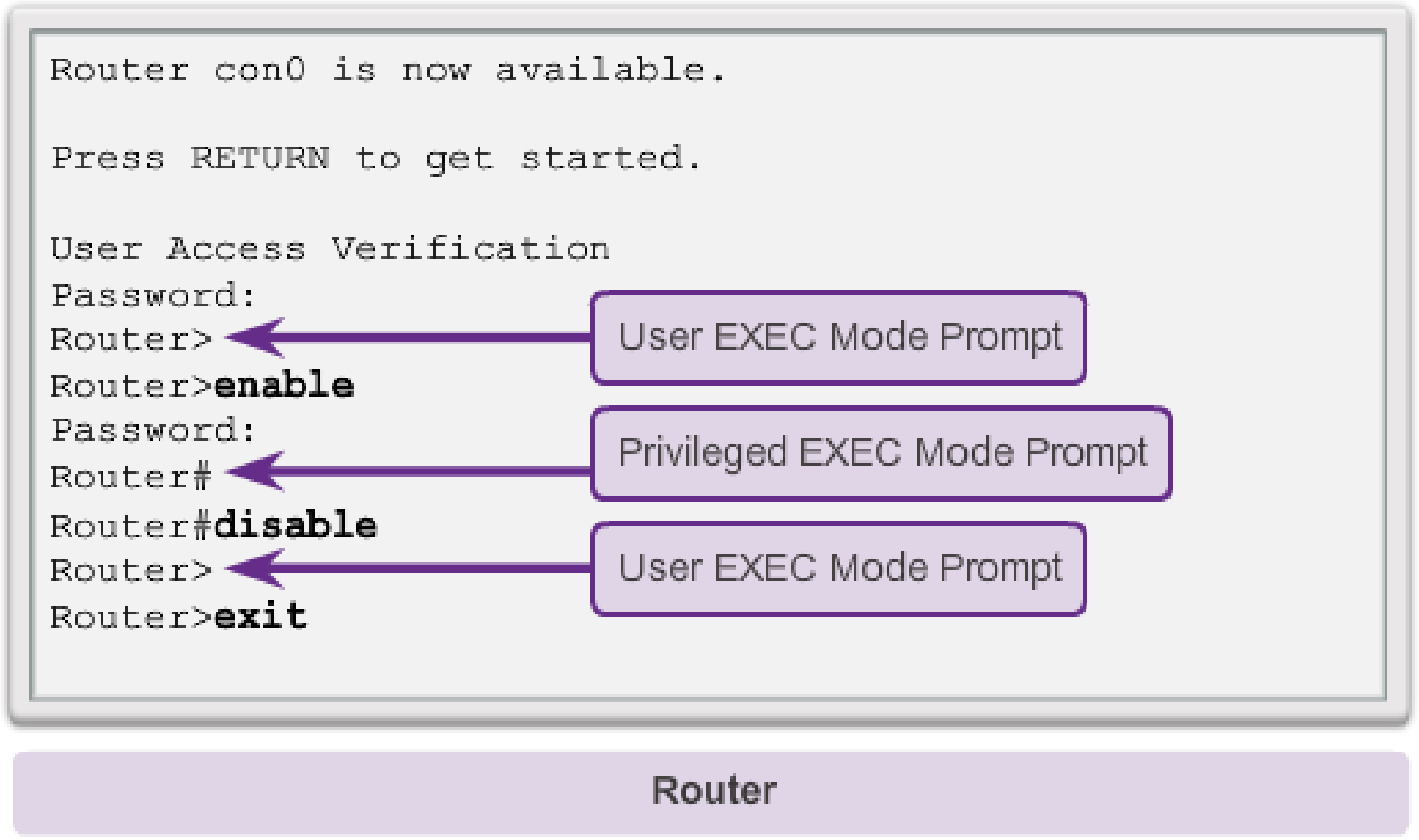
Switch#show running-config

Switch(config)#Interface FastEthernet 0/1

Switch(config-if)#Description connection to WEST LAN4
```


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Navigating Between IOS Modes



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Navigating Between IOS Modes (cont.)

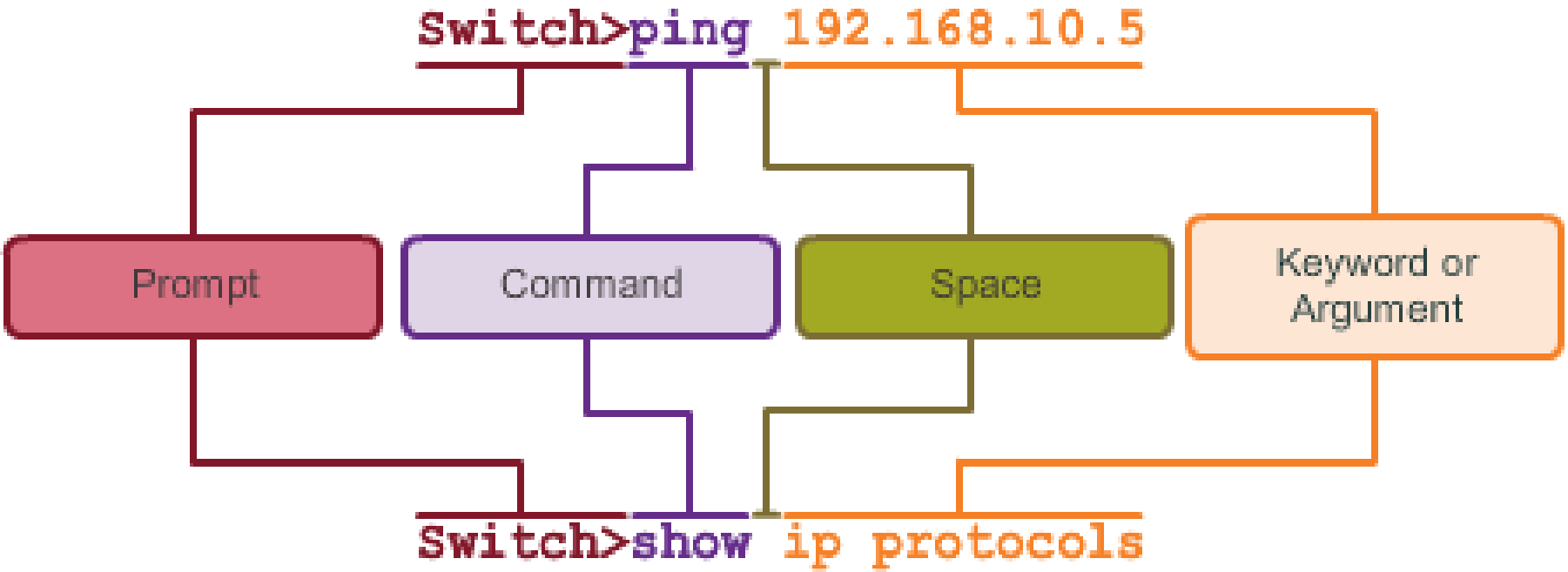
```
Switch> enable
Switch# configure terminal
Enter configuration commands, one per line.
End with CNTL/Z.
Switch(config)# interface vlan 1
Switch(config-if)# exit
Switch(config)# exit
Switch#
```

```
Switch# configure terminal
Enter configuration commands, one per line.
End with CNTL/Z.
Switch(config)# vlan 1
Switch(config-vlan)# end
Switch#
```

```
Switch# configure terminal
Enter configuration commands, one per line.
End with CNTL/Z.
Switch(config)# line vty 0 4
Switch(config-line)# interface fastethernet 0/1
Switch(config-if)# end
Switch#
```


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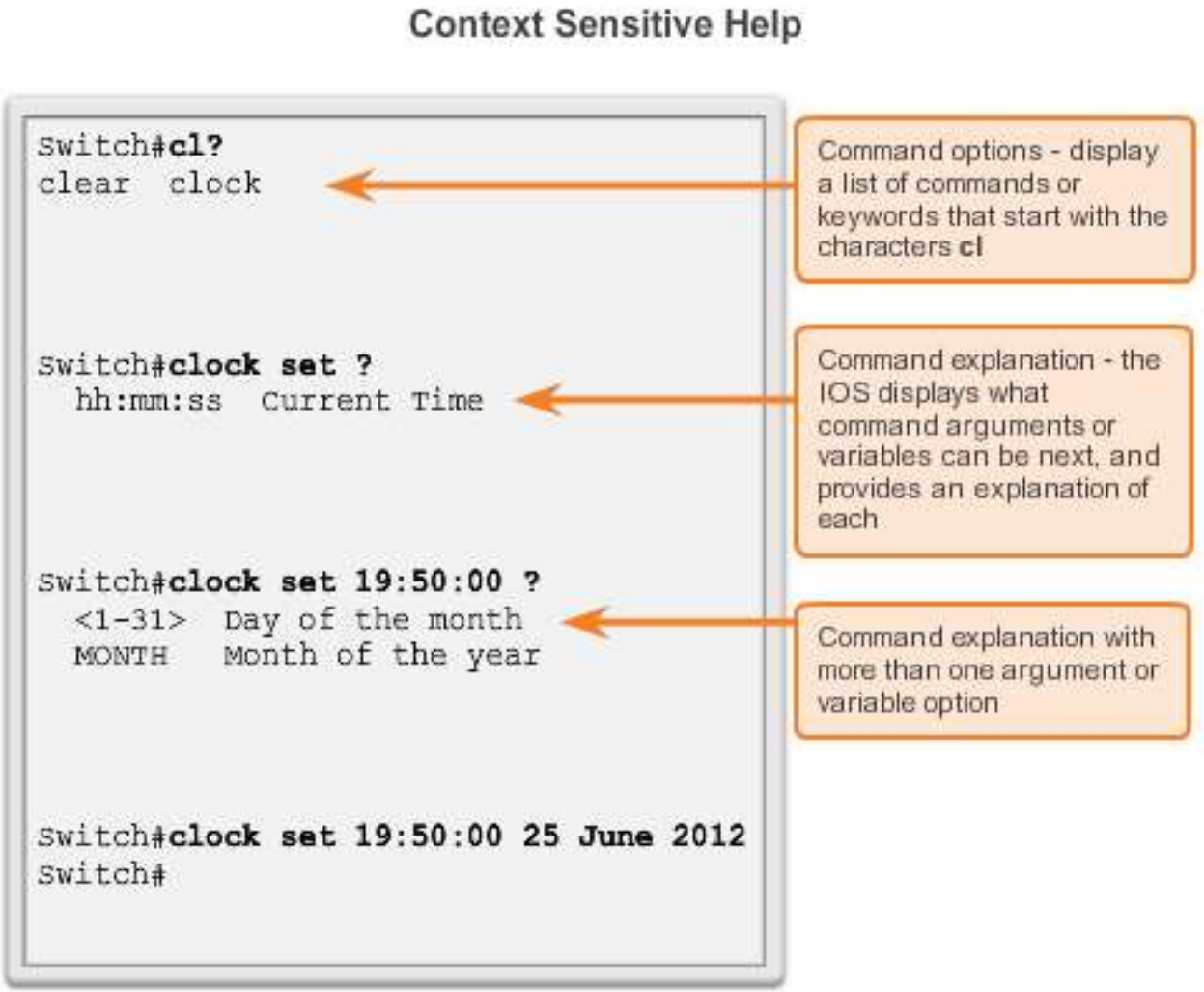
IOS Command Structure



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The Command Structure

Context-Sensitive Help



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Command Syntax Check

```
Switch#>clock set
% Incomplete command.
Switch#clock set 19:50:00
% Incomplete command.
```

The IOS returns a help message indicating that required keywords or arguments were left off the end of the command.

```
Switch#c
% Ambiguous command:'c'
```

The IOS returns a help message to indicate that there were not enough characters entered for the command interpreter to recognize the command.

```
Switch#clock set 19:50:00 25 6
                          ^
% Invalid input detected at '^'
marker.
```

The IOS returns a "^" to indicate where the command interpreter can not decipher the command.

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Hot Keys and Shortcuts

- **Tab –** Completes the remainder of a partially typed command or keyword.
- **Ctrl-R –** Redisplays a line.
- **Ctrl-A –** Moves to the beginning of the line.
- **Ctrl-Z –** Exits the configuration mode and returns to user EXEC.
- **Down Arrow –** Allows the user to scroll forward through former commands.
- **Up Arrow –** Allows the user to scroll backward through former commands.
- **Ctrl-shift-6 –** Allows the user to interrupt an IOS process such as **ping** or **traceroute**.
- **Ctrl-C –** Exits the current configuration or aborts the current command.

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The Command Structure

The show version Command

```
Router# show version
Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version
15.2(4)M1, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Thu 26-Jul-12 19:34 by prod_rel_team

ROM: System Bootstrap, Version 15.0(1r)M15, RELEASE SOFTWARE (fc1)

cisco1941 uptime is 41 minutes
System returned to ROM by power-on
System image file is ""flash0:c1900-universalk9-mz.SPA.152-
4.M1.bin""
Last reload type: Normal Reload
Last reload reason: power-on



This product contains cryptographic features and is subject to
United
States and local country laws governing import, export, transfer
and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use
encryption.
```

```
Router# show version
```


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2.2 Getting Basic



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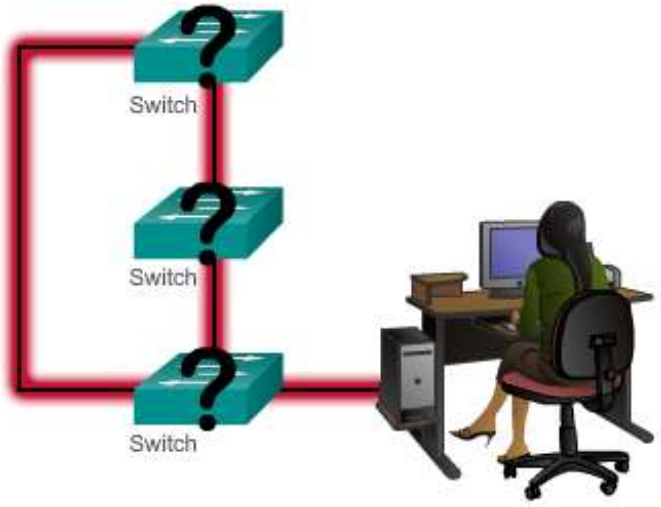
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Device Names

Some guidelines for naming conventions:

- Start with a letter
- Contains no spaces
- Ends with a letter or digit
- Uses only letters, digits, and dashes
- Be less than 64 characters in length

Without names, network devices are difficult to identify for configuration purposes.



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Configuring Hostnames

Configure a Hostname

Configure the switch hostname to be 'Sw-Floor-1'.

```
Switch# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#hostname Sw-Floor-1
Sw-Floor-1(config)#
```

You successfully configured the switch hostname.

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Securing Device Access

These are device access passwords:

- **enable password** – Limits access to the privileged EXEC mode
- **enable secret** – Encrypted, limits access to the privileged EXEC mode
- **console password** – Limits device access using the console connection
- **VTY password** – Limits device access over Telnet

Note: In most of the labs in this course, we will be using simple passwords such as **cisco** or **class**.

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Securing Privileged EXEC Access Mode

- Use the **enable secret** command, not the older **enable password** command.
- The **enable secret** command provides greater security because the password is encrypted.

```
Sw-Floor-1>enable
Sw-Floor-1#
Sw-Floor-1#conf terminal
Sw-Floor-1(config)#enable secret class
Sw-Floor-1(config)#exit
Sw-Floor-1#
Sw-Floor-1#disable
Sw-Floor-1>enable
Password:
Sw-Floor-1#
```


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Securing User EXEC Access

```
Sw-Floor-1(config)#line console 0
Sw-Floor-1(config-line)#password cisco
Sw-Floor-1(config-line)#login
Sw-Floor-1(config-line)#exit
Sw-Floor-1(config)#
Sw-Floor-1(config)#line vty 0 15
Sw-Floor-1(config-line)#password cisco
Sw-Floor-1(config-line)#login
Sw-Floor-1(config-line)#
```

- Console port must be secured; it reduces the chance of unauthorized personnel physically plugging a cable into the device and gaining device access.
- VTY lines allow access to a Cisco device via Telnet. The number of VTY lines supported varies with the type of device and the IOS version.

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Encrypting Password Display

Configuring Password Encryption

Enter the command to encrypt the plain text passwords.

```
Switch(config)# service password-encryption
```

Exit global configuration mode and view the running configuration.

```
Switch(config)# exit
Switch# show running-config
!
<output omitted>
!
line con 0
 password 7 094F471A1A0A
 login
!
line vty 0 4
 password 7 03095A0F034F38435B49150A1819
 login
!
!
end
```

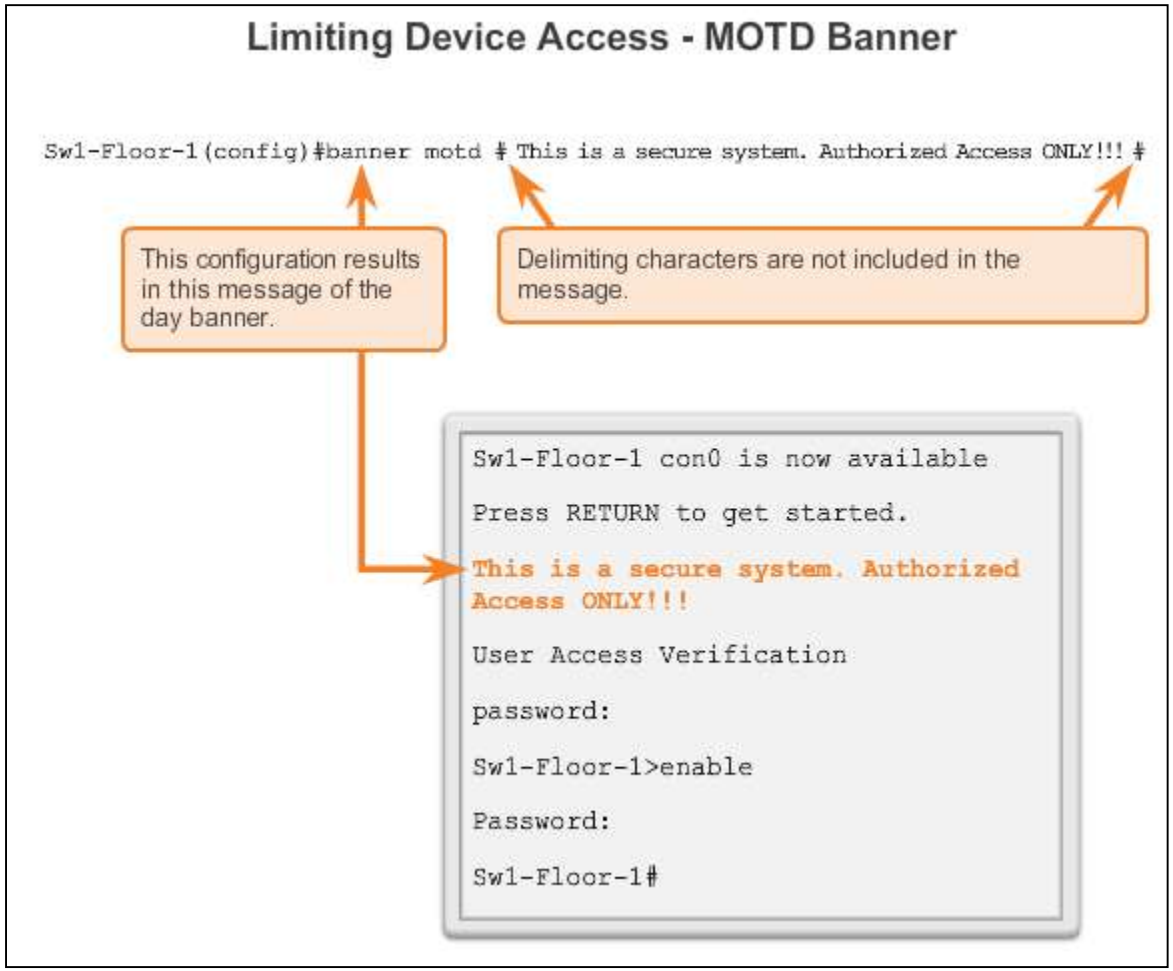
service password-encryption

- Prevents passwords from showing up as plain text when viewing the configuration
- Keeps unauthorized individuals from viewing passwords in the configuration file
- Once applied, removing the encryption service does not reverse the encryption

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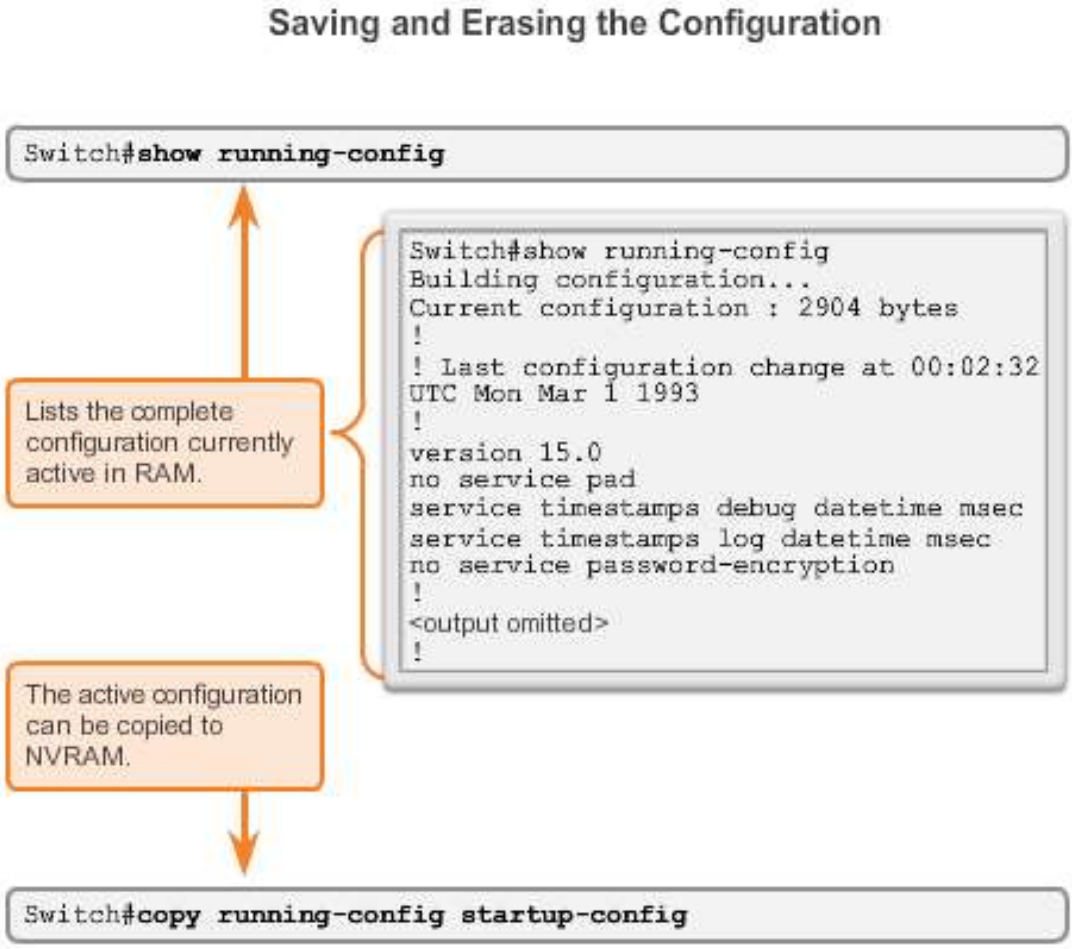
Banner Messages

- Important part of the legal process in the event that someone is prosecuted for breaking into a device
- Wording that implies that a login is "welcome" or "invited" is not appropriate
- Often used for legal notification because it is displayed to all connected terminals



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Configuration Files



- `Switch# reload`

  `System configuration has been modified. Save? [yes/no]: n`

  `Proceed with reload? [confirm]`

- Startup configuration is removed by using the **`erase startup-config`**

  `Switch# erase startup-config`

- On a switch, you must also issue the **`delete vlan.dat`**

  `Switch# delete vlan.dat`

  `Delete filename [vlan.dat]?`

  `Delete flash:vlan.dat? [confirm]`

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2.3 Addressing Schemes

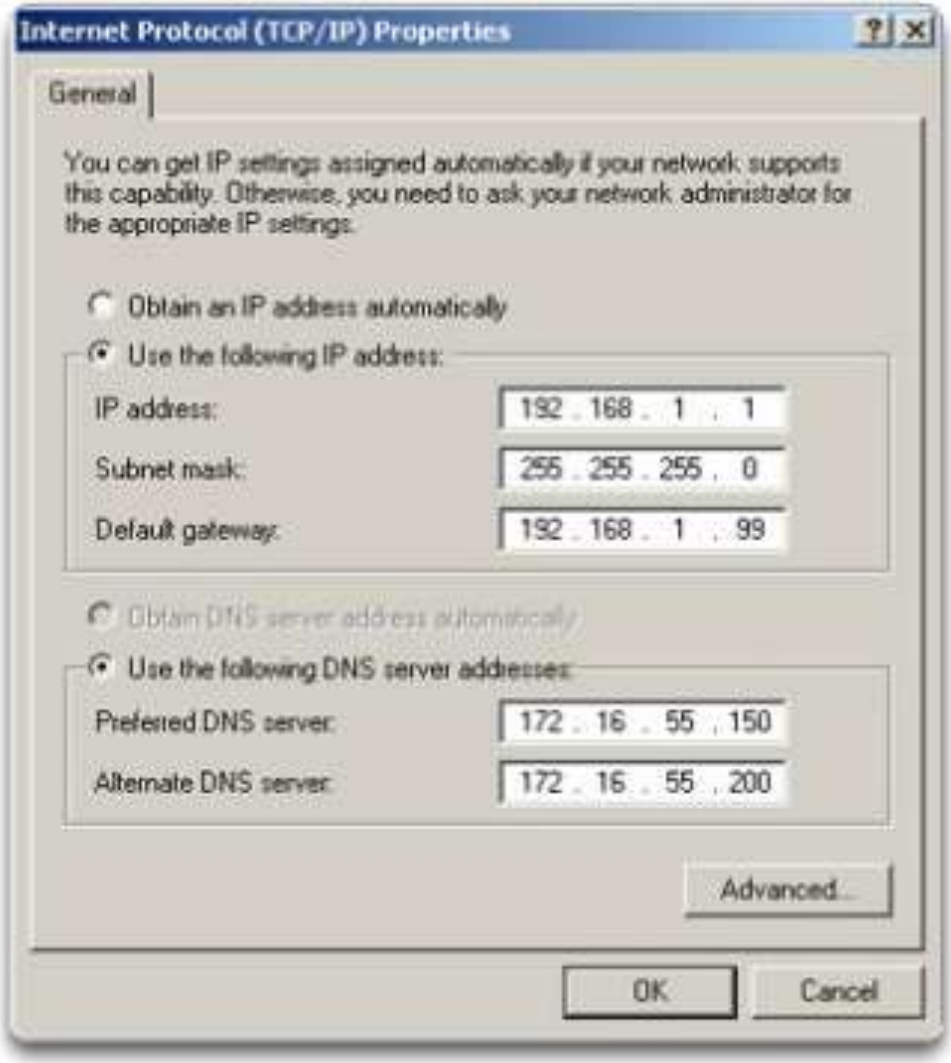


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IP Addressing of Devices

- Each end device on a network must be configured with an IP address.
- Structure of an IPv4 address is called *dotted decimal*.
- IP address displayed in decimal notation, with four decimal numbers between 0 and 255.
- With the IP address, a subnet mask is also necessary.
- IP addresses can be assigned to both physical ports and virtual interfaces.



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Configuring a Switch Virtual Interface

- **IP address** – Together with subnet mask, uniquely identifies end device on internetwork.
- **Subnet mask** – Determines which part of a larger network is used by an IP address.
- **`interface VLAN 1`** – Available in interface configuration mode,
- **`ip address 192.168.10.2 255.255.255.0`** – Configures the IP address and subnet mask for the switch.
- **no shutdown** – Administratively enables the interface.
- Switch still needs to have physical ports configured and VTY lines to enable remote management.

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Configuring a Switch Virtual Interface

Enter interface configuration mode for VLAN 1.

```
Switch(config)# interface vlan 1
```

Configure the IP address as '192.168.10.2' and the subnet mask as '255.255.255.0'.

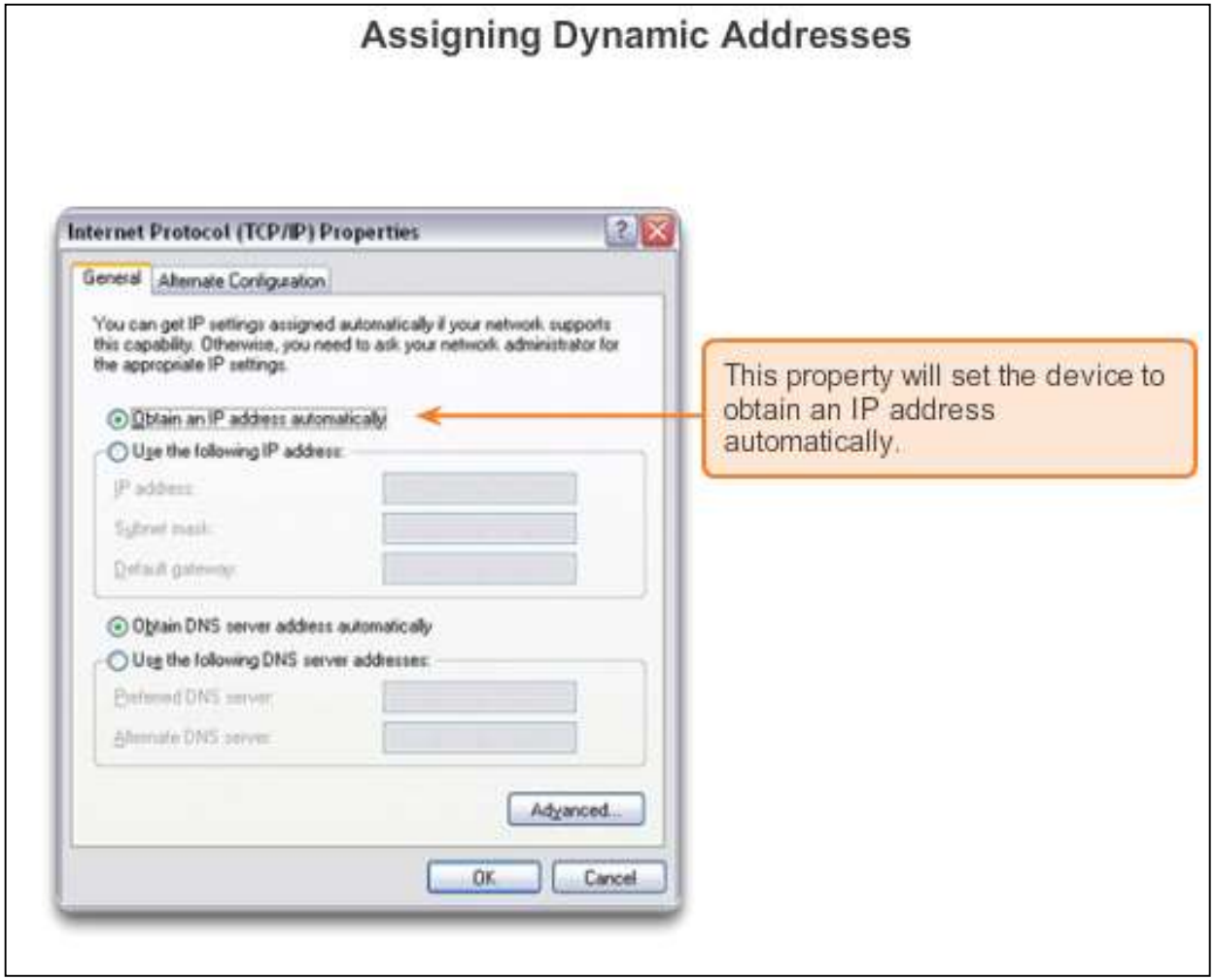
```
Switch(config-if)# ip address 192.168.10.2 255.255.255.0
```

Activate the interface.

```
Switch(config-if)# no shutdown
%LINK-5-CHANGED: Interface Vlan1, changed state to up
```

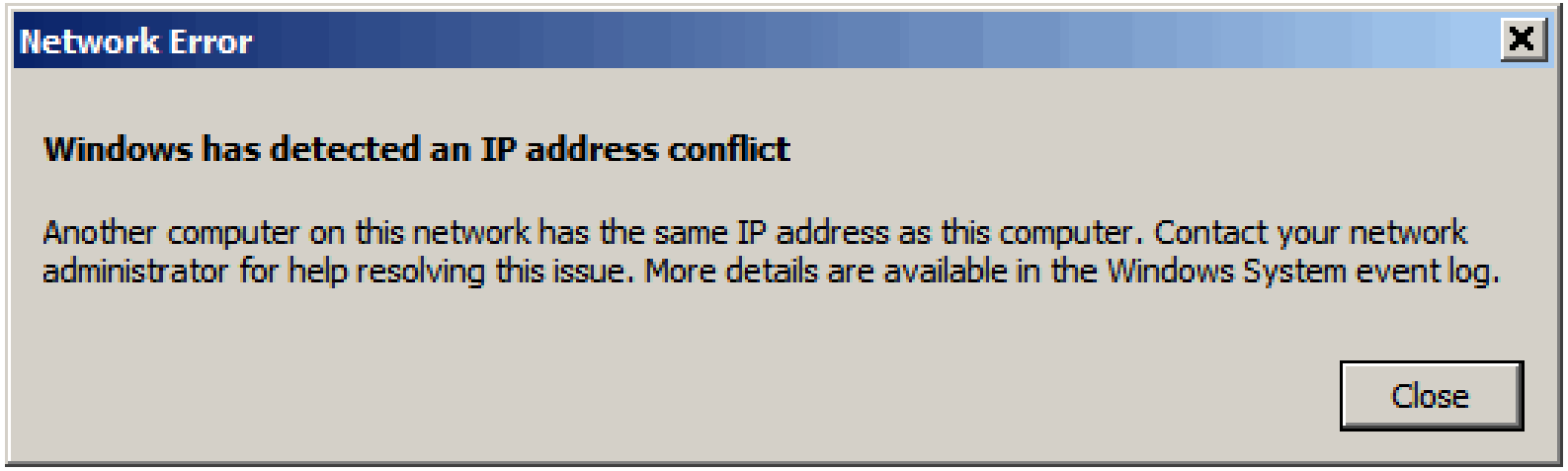

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Automatic IP Address Configuration for End Devices



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IP Address Conflicts



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Verifying Connectivity

Testing the Interface Assignment

Verifying the VLAN Interface Assignment

Enter the command to verify the interface configuration on S1.

```
S1# show ip interface brief
Interface          IP-Address      OK?  Method  Status      Protocol
FastEthernet0/1    unassigned      YES  manual  up          up
FastEthernet0/2    unassigned      YES  manual  up          up
<output omitted>
Vlan1              192.168.10.2    YES  manual  up          up
```

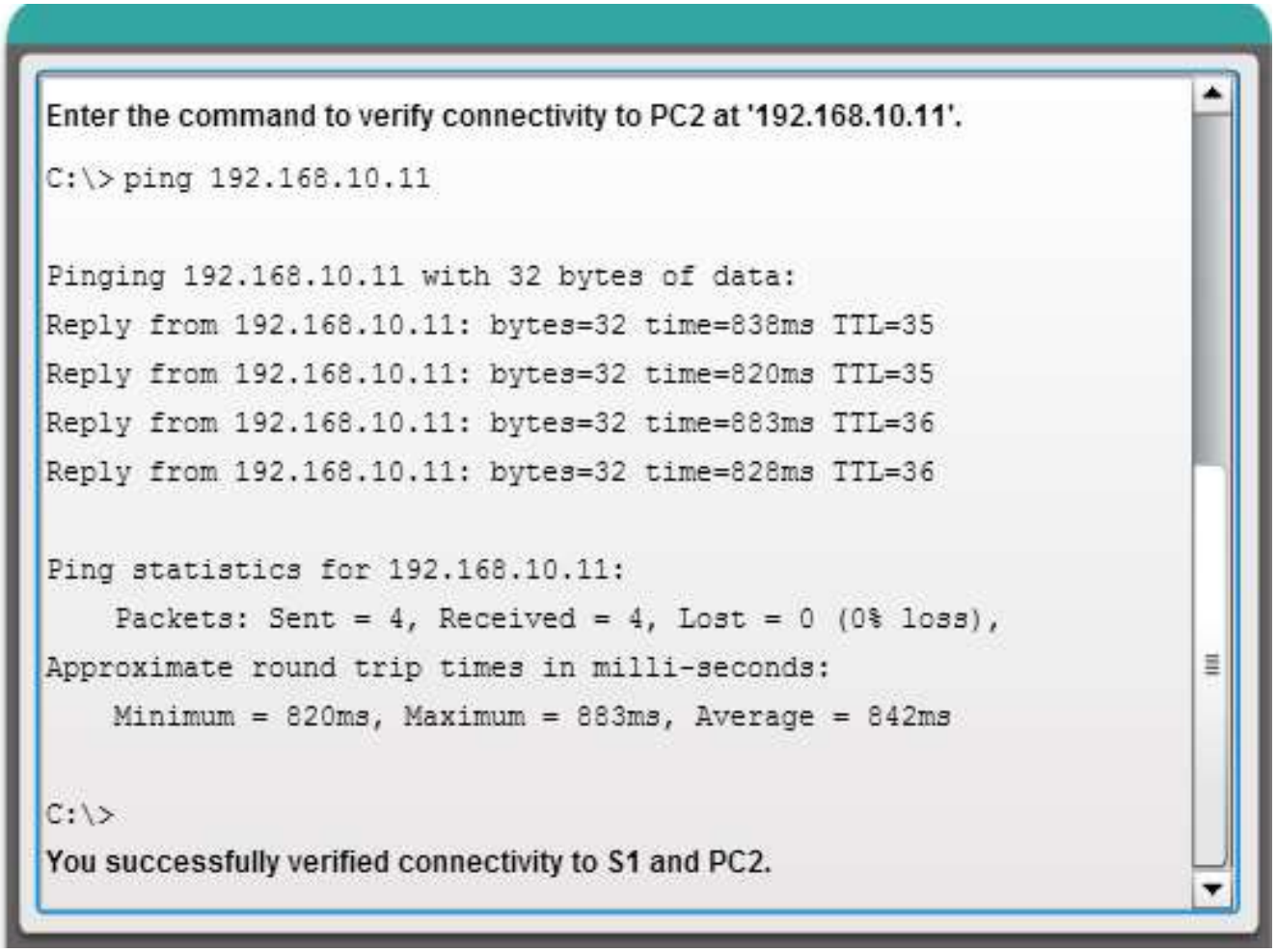
You are now on S2. Enter the command to verify the interface configuration on S2.

```
S2# show ip interface brief
Interface          IP-Address      OK?  Method  Status      Protocol
FastEthernet0/1    unassigned      YES  manual  up          up
FastEthernet0/2    unassigned      YES  manual  up          up
<output omitted>
Vlan1              192.168.10.3    YES  manual  up          up
```

You successfully verified the interface assignment on S1 and S2.

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Testing End-to-End Connectivity



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Chapter 2 Summary

Cisco IOS:

- The technician can enter commands to configure, or program, the device to perform various networking functions.
- Services are generally accessed using a command-line interface (CLI), which is accessed by either the console port, the AUX port, or through telnet or SSH.
- Once connected to the CLI, network technicians can make configuration changes to Cisco IOS devices.
- Cisco IOS is designed as a modal operating system, which means a network technician must navigate through various hierarchical modes of the IOS.
- Cisco IOS routers and switches support a similar modal operating system, support similar command structures, and support many of the same commands. In addition, both devices have identical initial configuration steps when implementing them in a network.

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Configuring a Network Operating System

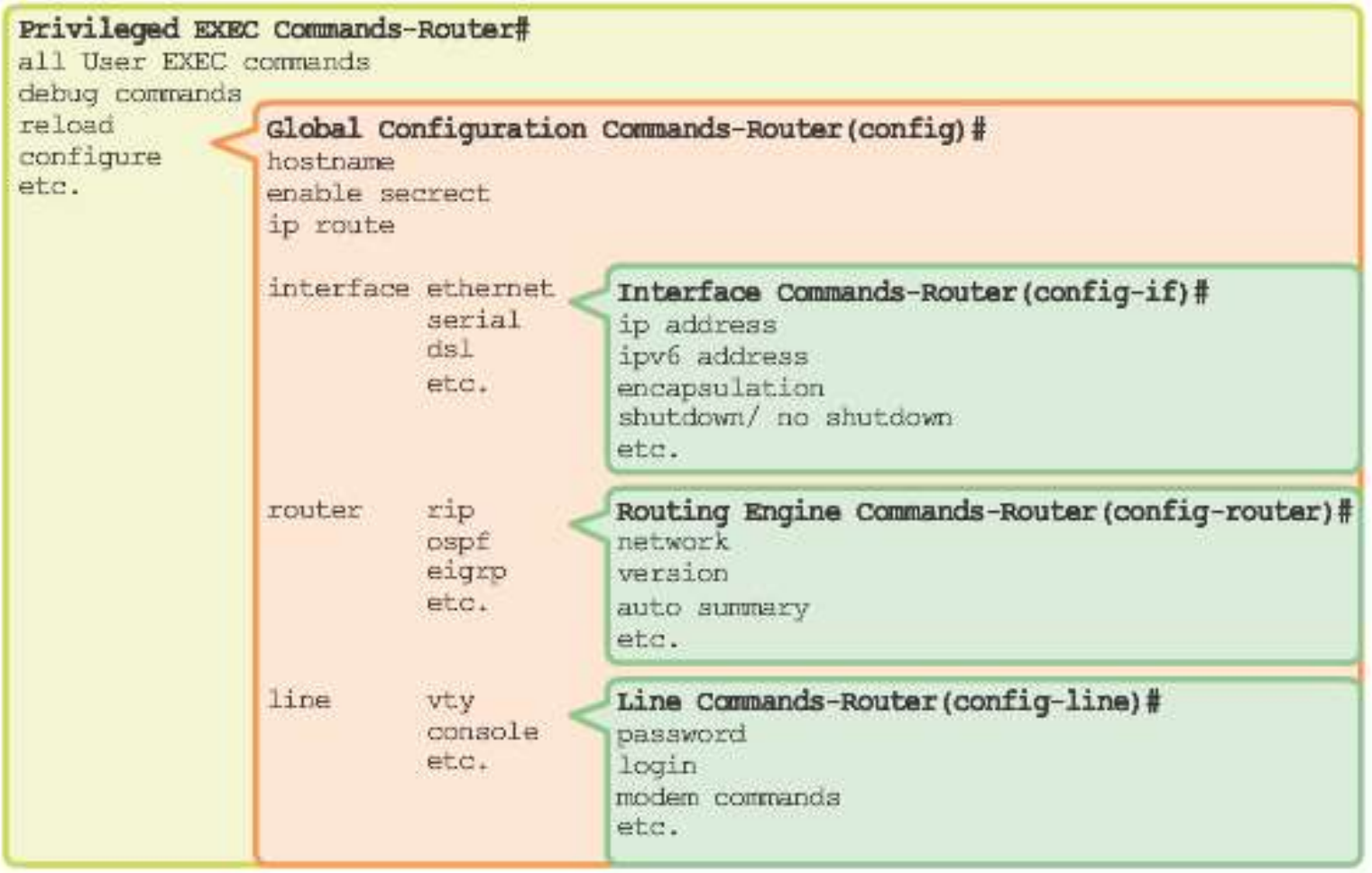
Chapter 2 Summary (cont.)

```
User EXEC Command-Router>
ping
show (limited)
enable
etc.
```


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Configuring a Network Operating System

Chapter 2 Summary (cont.)



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Homework

- Lab 2.1.4.6
- Lab 2.2.2.5
- Lab 2.3.2.4
- Lab 2.3.2.5
- Lab 2.4.1.1 (on board: commands and group them)
- Lab 2.4.1.2

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